

UBON RATCHATHANI AGROMET, Thailand

WMO: 484080

Lat: 15.233N

Lon: 105.033E

Elev: 426

StdP: 14.47

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	59.0	61.3	50.9	56.0	65.9	53.0	60.7	66.7	14.5	73.2	13.0	72.8	2.2	N/A	0.306

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	18.5	99.4	78.7	97.3	78.4	95.5	78.2	81.8	91.2	81.3	90.4	80.8	89.8	1.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.7	156.6	85.8	79.1	153.4	85.2	78.6	151.0	84.8	46.0	91.4	45.4	90.7	44.9	90.2	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.5	9.0	7.2	DB	55.2	102.1	2.2	2.7	53.6	104.1	52.3	105.7	51.1	107.2	49.5	109.2	
			WB	52.4	83.6	2.2	1.7	50.8	84.8	49.5	85.8	48.3	86.8	46.7	88.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.7	76.1	79.5	83.7	86.1	85.1	84.2	82.7	82.6	82.2	81.8	80.3	76.6
	DBStd	4.51	4.17	4.79	4.36	3.63	2.97	2.35	2.55	2.02	2.36	2.14	3.27	4.31
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11586	808	827	1045	1082	1089	1026	1014	1010	965	986	910	824
	CDD65	6112	344	407	580	632	624	576	549	545	515	521	460	359
	CDH74	64826	3080	4333	6785	7882	7325	6568	5780	5673	5151	4986	4223	3040
	CDH80	28087	1259	2140	3519	4195	3434	2756	2130	1978	1767	1896	1773	1240
(14) Wind	WSAvg	1.6	2.1	1.7	1.5	1.3	1.2	1.4	1.4	1.1	0.9	1.4	2.4	2.7
(15) Precipitation	PrecAvg	76.0	0.2	0.5	1.5	3.1	9.0	10.6	16.4	15.8	13.7	4.4	0.8	0.3
	PrecMax	91.7	1.4	1.9	4.1	6.4	15.7	18.6	27.5	28.0	29.0	11.7	2.3	3.4
	PrecMin	55.2	0.0	0.0	0.2	0.7	4.5	5.2	6.7	9.1	7.1	0.4	0.0	0.0
	PrecStd	10.8	0.4	0.5	1.1	1.8	2.9	3.6	5.4	4.5	5.0	2.5	0.7	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.4	97.7	100.9	103.8	99.9	96.8	95.4	93.1	93.4	93.3	94.5	94.3
		MCWB	74.5	75.9	77.8	78.5	79.4	80.4	79.8	81.0	81.2	79.1	78.5	76.3
	2%	DB	91.6	95.6	98.3	100.4	97.4	94.4	92.6	91.4	91.5	91.7	92.7	92.0
		MCWB	73.9	75.5	77.5	79.2	79.8	80.4	79.9	80.5	80.5	78.5	77.9	75.5
	5%	DB	89.4	93.6	96.3	97.8	95.2	92.5	90.7	89.9	89.9	90.4	91.1	89.9
		MCWB	73.1	75.0	77.1	78.9	79.9	80.1	79.6	80.0	80.0	78.2	77.2	74.7
	10%	DB	87.2	91.3	94.1	95.1	92.7	90.5	88.8	88.2	88.2	88.9	89.1	87.2
		MCWB	72.0	74.2	76.7	78.6	79.7	79.8	79.3	79.6	79.6	78.0	76.3	73.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.9	79.2	80.1	82.2	82.3	82.3	81.8	82.3	81.9	81.0	80.8	78.2
		MCDB	88.7	92.4	93.8	96.3	93.1	90.9	90.2	90.4	89.8	88.9	90.5	89.8
	2%	WB	76.3	78.2	79.2	81.0	81.4	81.4	80.8	81.4	81.4	80.1	79.5	76.9
		MCDB	86.9	90.9	93.1	94.1	91.9	90.2	89.2	89.0	88.9	88.0	89.2	88.8
	5%	WB	75.0	77.1	78.5	80.2	80.7	80.8	80.1	80.7	80.8	79.5	78.5	75.7
		MCDB	85.0	89.6	92.3	92.6	91.1	89.7	88.0	87.8	87.9	87.2	87.9	87.2
	10%	WB	73.7	75.9	77.8	79.5	80.1	80.2	79.6	80.0	80.1	78.8	77.6	74.4
		MCDB	84.2	87.7	91.0	91.3	89.8	88.6	87.0	86.5	86.6	86.2	86.2	85.2
(35) Mean Daily Temperature Range	5% DB	MDBR	22.9	23.8	20.9	18.5	15.4	13.2	11.8	11.4	11.7	14.4	17.4	20.7
		MCDBR	24.2	24.6	22.6	21.1	18.5	15.7	14.4	13.6	14.3	16.8	19.1	22.4
		MCWBR	8.6	7.9	6.8	5.5	4.9	4.7	4.9	5.0	5.2	6.0	6.6	8.1
	5% WB	MCDBR	21.0	21.1	20.1	18.5	16.1	14.1	13.0	12.7	13.5	14.7	17.1	20.9
		MCWBR	8.2	7.0	6.5	5.5	5.3	5.1	5.1	5.2	5.3	5.7	6.4	8.0
(40) Clear-Sky Solar Irradiance	taub		0.458	0.581	0.662	0.610	0.498	0.465	0.461	0.461	0.492	0.500	0.447	0.438
	taud		2.168	1.832	1.691	1.854	2.195	2.303	2.313	2.305	2.208	2.159	2.294	2.267
	Ebn at Noon		264	236	221	233	257	263	265	267	260	255	266	266
	Edh at Noon		46	67	79	67	47	42	42	42	46	48	40	41
(44) All-Sky Solar Radiation	RadAvg		1634	1722	1689	1764	1731	1594	1496	1437	1446	1589	1583	1552
(45)	RadStd		80	78	57	79	87	89	133	101	138	118	67	59

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	+0.52	N/A	N/A	N/A	N/A	+0.40	+0.36	N/A	N/A	+185	+184